

RESEARCH ARTICLE

Functional Impact of 14 Single Nucleotide Polymorphisms Causing Missense Mutations of Human $\alpha 7$ Nicotinic Receptor

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Abstract

The $\alpha 7$ nicotinic receptor (nAChR) is a major subtype of the nAChRs in the central nervous system, and the receptor plays an important role in brain function. In the dbSNP database, there are 55 single nucleotide polymorphisms (SNPs) that cause missense mutations of the human $\alpha 7$ nAChR in the coding region. In this study, we tested the impact of 14 SNPs that cause missense mutations in the agonist binding site or the coupling region between binding site and channel gate on the receptor function. The wild type or mutant receptors were expressed or co-expressed in *Xenopus* oocytes, and the agonist-induced currents were tested using two-electrode voltage clamp. Our results demonstrated that 6 mutants were nonfunctional, 4 mutants had reduced current expression, and 1 mutants altered ACh and nicotine efficacy in the opposite direction, and one additional mutant had slightly reduced agonist sensitivity. Interestingly, the function of most of these nonfunctional mutants could be rescued by $\alpha 7$ nAChR positive allosteric modulator PNU-120596 and agonist-PAM 4BP-TQS. Finally, when coexpressed with the wild type, the nonfunctional mutants could also influence the receptor function. These changes of the receptor properties by the mutations could potentially have an impact on the physiological function of the $\alpha 7$ nAChR-mediated cholinergic synaptic transmission and anti-inflammatory effects in the human SNP carriers. Rescuing the nonfunctional mutants could provide a novel way to treat the related disorders.

Introduction

Cholinergic transmission plays an important role in brain function, such as learning and memory through the neurotransmitter acetylcholine and its receptors [1]. nAChRs are acetylcholine-operated ion channels. They belong to the pentameric ligand-gated ion channel superfamily, which includes vertebrate cation-selective nicotinic receptors [2], serotonin receptor type 3 [3] and zinc-activated ion channel [4], and anion-selective GABA_{A/C} receptors [5–7] and glycine receptors [8]. The pentameric ligand-gated ion channels are allosteric proteins [9], in which, the orthosteric ligand binding sites are located in the extracellular N-terminal domain, and the ion conducting channel is formed by the transmembrane domain. The binding site is formed by 6 binding loops. Loops A, B, C from one subunit forms the principal side of the binding pocket, whereas loops D, E, F from the neighboring subunit form the complementary side of the binding pocket [10]. Thus, neurotransmitter binding to the N-terminal binding site can allosterically (remotely) control the channel gate through an evolutionarily interconnected allosteric network [11, 12]. There are 17 subunits and isoforms of nAChRs: $\alpha 1$ – $\alpha 10$, $\beta 1$ – $\beta 4$, γ , δ , and ϵ [1, 2, 13]. The pentameric receptor can have five identical (such as in $\alpha 7$ nAChR) or different (such as in $\alpha 4\beta 2$ nAChR) subunits in a receptor. In the mammalian brain, the major subtypes are heteromeric $\alpha 4\beta 2$ and homomeric $\alpha 7$ nAChRs [2].

Homomeric $\alpha 7$ nAChR is widely distributed in the CNS and autonomic ganglia. It has much higher calcium permeability than other subtypes of nAChRs [1]. In addition to excitatory synaptic transmission, $\alpha 7$ nAChR also plays an important role in transmitter release [14], neurite growth [15], neuronal survival and apoptosis [16] and neuronal plasticity [17]. Dysfunction of $\alpha 7$ nAChR is associated several neurological and psychiatric disorders, such as schizophrenia [18]. In addition, $\alpha 7$ nAChR gene deletion is associated with seizures and mental retardation [19, 20].

Transcript of $\alpha 7$ nAChR subunit is also found in immune cells, especially in macrophages in peripheral blood and microglia cells in the CNS [21]. In fact, nicotine has anti-inflammatory effects, which can be blocked by $\alpha 7$ selective antagonist [22]. Experimental evidence suggests that there are cholinergic anti-inflammatory pathways in peripheral and CNS via $\alpha 7$ nAChR expressed on macrophage or microglia [23, 24]. These studies suggest the essential role of $\alpha 7$ nAChR in inhibiting cytokine synthesis by the cholinergic anti-inflammatory pathway. A more recent study suggests that the $\alpha 7$ nAChR mediates the inhibition of T cell function through the regulatory T cells [25].

Single Nucleotide Polymorphisms (SNPs) are single base variations of the DNA sequence among individuals of the same species [26]. In the coding region of exons, a SNP can cause missense mutations, which result in substitution of an amino acid residue at the protein level, which may have impact on protein expression and/or function.

The SNP database [27], dbSNP, in the National Center for Biotechnology Information (NCBI) listed 55 SNPs causing missense mutations of the human $\alpha 7$ nAChR. Mapping these mutations in the 3D structure of the $\alpha 7$ nAChR subunit homology model, we identified 14 $\alpha 7$ nAChR SNPs that cause missense mutations in the agonist binding loops or the coupling region between amino terminal domain and transmembrane domain (Fig 1). Specifically, these mutants are located in the binding loop A (Y93C), loop C (C191Y, K192R, and D197N), loop D (W55G); coupling loop 2 (N47D), loop 9 (N171S and E173K), pre-M1 (R205C, R205H, and R206C), and extracellular end of M1 (Y211C, G212V and G212S). In this study, we characterized the functional impact of these 14 mutants for the neurotransmitter acetylcholine, and smoking related agonist nicotine. These SNPs and corresponding mutations are listed in Table 1. It should be pointed out that although in this study we selected the mutants in the binding site and coupling region, we cannot exclude possibility that mutations in other part of

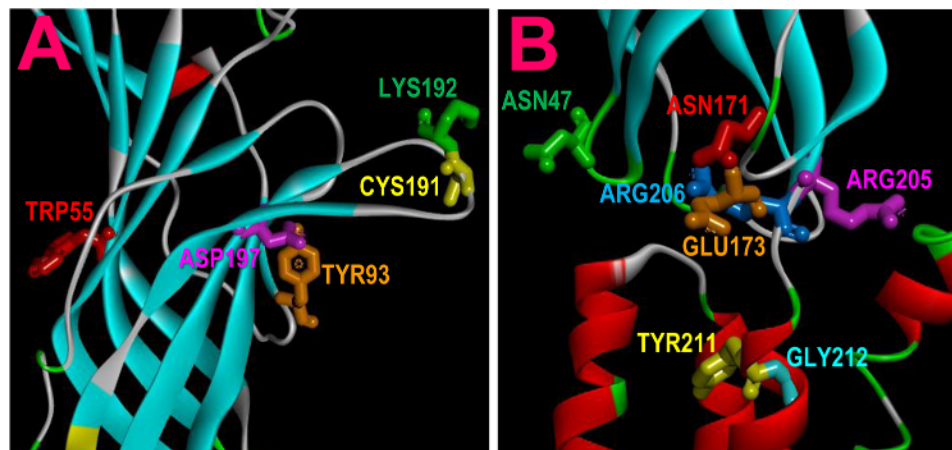


Fig 1. Location of 14 mutations in the $\alpha 7$ nAChR receptor. A: The residues with the mutations in the binding region (loops A (TYR93), and C (CYS191, LYS192, ASP197) of the principal side of a binding pocket, and D (TRP55) of the complementary side of the neighboring binding pocket); B: The residues with the mutations in the coupling region (loop 2 (ASN47), loop 9 (ASN171 and GLU173), pre-M1 (ARG205 and ARG206) and M1 (TYR211 and GLY212)).

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the receptor can influence channel function. As demonstrated by single channel studies to map gating energy changes upon mutagenic perturbation in multiple domains, although sources of gating energy are mainly from the binding loops C, B, and A, other part of the receptor can also influence channel gating, but to a lesser extent. [28, 29].

Materials and Methods

Mutagenesis and cRNA Preparation

The cDNA encoding wild type human $\alpha 7$ nAChR subunit was cloned into pGEMHE oocyte expression vector with T7 orientation. The mutations were made with the PCR-based Quik-Change method of the site-directed mutagenesis (Stratagene, Hercules, CA, USA) with Phusion DNA polymerase (New England Biolab, Ipswich, MA). The mutations were confirmed by

Table 1. List of the SNPs of the human $\alpha 7$ nAChR in this study.

SNP name	Missense mutation	Numbering w/o signal peptide	Location in the receptor
rs201473594	N69D	N47D	Loop 2 (coupling)
rs12899798	W77G	W55G	Loop D (binding)
rs200908085	Y115C	Y93C	Loop A (binding)
rs201108331	N193S	N171S	Loop 9 (coupling)
rs201210785	E195K	E173K	Loop 9 (coupling)
rs200236230	C213Y	C191Y	Loop C (binding)
rs143167432	K214R	K192R	Loop C (binding)
rs377100778	D219N	D197N	Loop C (binding)
rs140316734	R227C	R205C	Pre-M1 (coupling)
rs138222088	R227H	R205H	Pre-M1 (coupling)
rs201224804	R228C	R206C	Pre-M1 (coupling)
rs142728508	Y233C	Y211C	M1 (coupling)
rs201524804	G234S	G212S	M1 (coupling)
rs377388459	G234V	G212V	M1 (coupling)

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automated DNA sequencing of the entire coding region. The wild type and mutant cDNAs were amplified by PCR with Phusion DNA polymerase and M13 forward and reverse primers, and used as the DNA templates for cRNA synthesis. The cRNAs were transcribed by standard *in vitro* transcription using T7 RNA polymerase. After degradation of the DNA template by RNase-free DNase I, the cRNAs were purified and resuspended in diethyl pyrocarbonate (DEPC)-treated water. cRNA yield and integrity were examined with an Eppendorf BioPhotometer and a 1% agarose gel.

Oocyte preparation and injection

Oocytes were harvested from female *Xenopus laevis* (Xenopus I, Ann Arbor, MI, USA), using the protocol "Xenopus Care and Use" approved by the Institutional Animal Care and Use Committee of the St. Joseph's Hospital and Medical Center for this study. Briefly, the frog was anesthetized by 0.2% MS-222. The ovarian lobes were surgically removed and placed in the incubation solution consisting of (in mM) 82.5 NaCl, 2.5 KCl, 1 MgCl₂, 1 CaCl₂, 1 Na₂HPO₄, 0.6 theophylline, 2.5 sodium pyruvate, and 5 HEPES; 50 U/ml penicillin, and 50 μ g/ml streptomycin, pH 7.5. The frog was given analgesic xylazine hydrochloride (10mg/kg, ip) and antibiotic gentamicin (ip, 3mg/kg) after surgery, and then allowed to recover from the surgery before being returned to the incubation tank. The animals were euthanized under anesthesia with MS-222 after the third surgery. The ovarian lobes were cut into small pieces and digested with 1 Wunsch unit/ml liberase blendzyme 3 (Roche Applied Science, Indianapolis, IN, USA) with constant stirring at room temperature for 1.5–2 hours. The dispersed oocytes were thoroughly rinsed with the above solution. The stage VI oocytes were selected and incubated at 16°C before injection. Micropipettes for injection were pulled from borosilicate glass on a Sutter P87 horizontal puller, and the tips were cut with forceps to ≈ 40 μ m in diameter. The cRNA was drawn up into the micropipette and injected into oocytes with a Nanoject micro-injection system (Drummond, Broomall, PA, USA) at a total volume of 20–60 nl.

Two-electrode voltage-clamp

Two to 4 days after injection, the oocytes expressing the wild type or mutant receptors were placed in a custom made small volume chamber with continuous perfusion with the calcium-free oocyte Ringer's solution, which consisted of (in mM) 92.5 NaCl, 2.5 KCl, 1.8 BaCl₂, and 5 HEPES, pH 7.5. The chamber was grounded through an agar KCl bridge. The oocytes were voltage-clamped at -70 mV to measure ACh-, nicotine-, or 4BP-TQS-induced currents using an AxoClamp 900A amplifier (Molecular Devices, Sunnyvale, CA, USA). The current signal was filtered at 50 Hz with the built-in 4 pole low-pass Bessel filter in the AxoClamp 900A and digitized at 100 Hz with Digidata1440A (Molecular Devices).

Drug Preparation

Acetylcholine chloride and nicotine (SigmaAldrich, St. Louis, MO, USA) PNU-120596 and 4BP-TQS (Tocris, Bristol, UK), atropine (SigmaAldrich) stock solutions were prepared from the solid and stored at -20°C in aliquots. Working concentrations of these drugs were prepared from stock solutions immediately before use.

Data Analysis

The peak current and net charge of the ACh-induced current were measured using Clampfit10.3. The peak current was measured relative the average baseline within 1 second before ACh application switching. The net charge (the area under the curve) was measured with the

start point 0.5 second after the pinch valve switching (~0.5 second before inward current rising) to 4 seconds later, when the fast decay phase of the currents induced by all concentrations of ACh was essentially completed for the wild type and mutants, using the average baseline within 1 second before agonist application switching. The concentration-response relationship of the agonist-induced current or charge in recombinant nAChRs was least-squares fit to a Hill equation with GraphPad Prism 6.0 (GraphPad Software, Inc, La Jolla, CA) to derive the EC_{50} (the agonist concentration required for inducing a half maximal change), Hill coefficient (the slope factor), and maximum current, which was then used to normalize the concentration-response curve from individual oocytes. The average of the normalized currents or charges for each agonist concentration was used to plot the data. All the data were presented as mean $\pm$ SEM (standard error). Statistical comparisons for $\log EC_{50}$ s, agonist-induced currents at a saturation concentration, or allosteric modulator rescued currents between wild type/blank control and multiple mutants were performed with one way ANOVA, or two-sided grouped t-test for two group comparison. Correlation between two parameters was performed by linear regression analysis in Prism6.0.

Homology Modeling

The homology model of the human $\alpha 7$ nAChR subunit was made with ICM Pro 3.7–2C (Mol-Soft, San Diego, CA) using chain A of the EM structure of *Torpedo* nicotinic receptor (PDBID: 2BG9 chain A) as the template for Fig 1. The resulting models were used to map the SNPs in the 3D structure using Discovery Studio 4.0 Visualizer (Biovia, San Diego, CA). The same software was also used to present the crystal structures of acetylcholine binding protein (AChBP) with ACh or nicotine in the binding pocket (PDBID: ACh binding with AChBP: 3WIP, nicotine binding with AChBP: 1UW6).

Results

Influence of mutations on the currents induced by ACh and nicotine

To test the impact of the 13 mutants (excluding W55G, which was tested in a separate experiment, as it was reported in the 2010 Society for Neuroscience Annual Meeting), we injected the same amount of cRNAs of the wild type and 13 mutants into *Xenopus* oocytes and tested receptor function with 3.16 mM ACh, a saturation concentration of the wild type $\alpha 7$ nAChR. Fig 2A shows the average currents induced by ACh for each construct. Note that except for two G212 mutants and K192R, all the remaining mutants exhibited reduced currents when compared to the wild type. Among them, R205C, R205H, N171S, and N47D mutants exhibited 3.5- to 5-fold reduction. But their current levels were still significantly higher than the un-injected control. Interestingly, Y93C, E173K, D197N, C191Y, R206C, and Y211C were nearly insensitive to 3.16 mM ACh. The very low ACh-induced currents are likely due to muscarinic receptor activation by the high ACh concentration, despite the presence of 1 μ M atropine, because these small currents were not statistically different from the current induced by 3.16 mM ACh in the un-injected oocytes. Thus, these mutants are essentially nonfunctional. The responses to 3.16mM nicotine are also shown in Fig 2A. The current responses to nicotine for all constructs exhibited a pattern similar to that for ACh responses. High concentration nicotine-induced small currents in the same set of six mutants were also not significantly different from that from the blank oocytes. Due to higher background current induced by 3.16mM nicotine in un-injected control and lower nicotine-induced current, the nicotine-induced currents in R205C, R205H, N171S, and N47D mutants did not reach statistical significance when compared to the un-injected control. The mechanism for the nonspecific effect of high concentration nicotine on the oocytes is unknown. However, we have noticed that it is oocyte batch dependent. In